

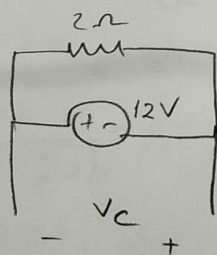
1) RLC Circuits:

- a) ① V_C cannot change instantaneously
 $V_C(0^+) = V_C(0^-) = 0V$
 ② i_L cannot change instantaneously
 $i_L(0^+) = i_L(0^-) = 0A$
 ③ At $t=0$ voltage source steps to $80V$
 $V_R(0^+) = 80V$

b) ① $V_L = L \frac{di_L}{dt}$ $V_L(0^+) = V_{source} = 80V$
 $\frac{di_L(0^+)}{dt} = \frac{V_L(0^+)}{L} = \frac{80}{2 \times 10^{-3}} = 40kA/s$
 ② $i_C = C \frac{dV_C}{dt}$ $i_C(0^+) = i_R(0^+) = 3.2mA$
 $\frac{di_C(0^+)}{dt} = 40kA/s$

c) ① $i_R(\infty) = \frac{V_{source}}{R_1 + R_2} = \frac{80}{25k + 20k} = \frac{80}{45k} = 1.78mA$
 ② $i_L(\infty) = i_R(\infty) = 1.78mA$ ③ $i_C(\infty) = 0A$

② for $t < 0$



$V_0 = -12V$

$i_0 = 6A$

for $t > 0$;

$\alpha = 2RL = 2/(2 \cdot 0.5) = 2$

$\omega_0 = \frac{1}{\sqrt{LC}} = 2\sqrt{2}$

$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{8 - 4} = 2$

$i(t) = (A \cos 2t + B \sin 2t) e^{-2t}$

$i(0) = 6 = A$

$\frac{di}{dt} = -2(6 \cos 2t + B \sin 2t) e^{-2t} + (-12 \sin 2t + 2B \cos 2t) e^{-2t}$
 $di(0)/dt = -12 + 2B$
 $= -(1/L) [Ri(0) + V_C(0)] = -2(0)20$
 $B = 6$ $i(t) = (6 \cos 2t + 6 \sin 2t) e^{-2t} A$

③

$$v(0^+) = v_0 = 0V$$

$$L \frac{d^2 v(t)}{dt^2} + R \frac{dv(t)}{dt} + \frac{v(t)}{C} = v_s$$

$$3 \frac{d^2 v(t)}{dt^2} + 46 \frac{dv(t)}{dt} + 27v(t) = 24 \quad (\text{standard form})$$

$$\frac{d^2 v(t)}{dt^2} + \frac{46}{3} \frac{dv(t)}{dt} + 9v(t) = 0 \quad (\text{homogeneous form})$$

$$v_h(t) = e^{st}$$

$$3s^2 + 46s + 27 = 0$$

$$s_1 = -0.611 \quad s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \Rightarrow s = \frac{-46 \pm 42.33}{6}$$

$$s_2 = -14.055$$

$$v_p = K \quad 3(0) + 46(0) + 9K = 24 \quad v_h(t) = Ae^{-0.611t} + Be^{-14.055t}$$

$$K = \frac{8}{9} \quad v_p = \frac{8}{9}$$

$$v(t) = v_h(t) + v_p(t)$$

$$v(t) = Ae^{-0.611t} + Be^{-14.055t} + \frac{8}{9}$$

$$v(t) = Ae^{-0.611t} + Be^{-14.055t} + \frac{8}{9}$$

$$A+B = -\frac{8}{9}$$

$$v(0) = 0$$

④ for $t=0$ $i(0)=0$ $v_c(0) = \frac{40 \cdot 30}{50} = 24V$

$$t > 0$$

$$\alpha = \frac{R}{2L} = \frac{(40+60)}{5} = 20 \quad \omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \cdot 2.5}} = 20$$

$$\omega_0 = \alpha$$

$$i(t) = [(A+Bt)e^{-20t}] \quad i_0 = 0 = A$$

$$di/dt = [Be^{-20t}] + [-20(Bt)e^{-20t}]$$

$$di(0)/dt = -(1/L) [Ri(0) + v_c(0)] = -\frac{1}{2.5} \cdot 24$$

$$B = -9.6 \quad i(t) = -9.6te^{-20t} \text{ A}$$

$$\textcircled{5} \quad t=0^- \quad v_0(0) = (81(10) \cdot 30 = 24 \text{ V}$$

for $t > 0$

$$\alpha = \frac{1}{2RC} = \frac{1}{4}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{\frac{1}{4}}} = 2$$

for $\alpha < \omega_0$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{4 - \frac{1}{16}} = 1,98$$

$$v_0(t) = (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t}$$

$$v_0(0) = 24 \text{ V} = A_1$$

$$t=0 \Rightarrow i_0(t) = 0$$

$$dv_0/dt = -\alpha (A_2 \sin \omega_d t + A_1 \cos \omega_d t) + (-\omega_d A_1 \sin \omega_d t + \omega_d A_2 \cos \omega_d t) e^{-\alpha t}$$

$$t=0 \quad dv_0(0)/dt = 0 = -\alpha A_1 + \omega_d A_2$$

$$A_2 = (\alpha / \omega_d) A_1 = \frac{1}{4} \cdot 24 \cdot 1,98 = 3,02$$

$$v_0(t) = 24 \cos(1,98t) + 3,02 \sin(1,98t) e^{-t/4} \text{ V}$$

$$i_0(t) = 0,000131 \cos(1,98t) - 12,095 \sin(1,98t) e^{-t/4} \text{ A}$$

⑥

$$i_L(0^+) = i_L(0^-) = I_0$$

$$i_L(0^-) = I_0 = 4A \quad i_L(0^+) = 4A$$

$$V_L = L \frac{di}{dt} \quad V_R = iR \quad V_C = \frac{1}{C} \int i dt$$

$$12 - 16i - 2i - L \frac{di}{dt} = 0$$

$$12 - 16i - 2 \frac{di}{dt} = 0$$

$$\frac{di}{dt} + 8i = 6$$

$$i(t) = i_p + i_h \quad \begin{array}{l} p = \text{particular} \\ h = \text{homogenous} \end{array}$$

i_h

$$\frac{di}{dt} = -8i$$

$$\ln|i| = -8t + K \Rightarrow i_h(t) = K e^{-8t}$$

i_p

$$\frac{d(i_p)}{dt} + 8i_p = 6 \Rightarrow 8i_p = 6 \quad i_p = 0,75A$$

$$i(t) = i_h + i_p = (K e^{-8t} + 0,75)A$$

$$\text{at } t=0 \quad i(0) = i_L(0^+) = 4A$$

$$4 = K e^{-8(0)} + 0,75$$

$$K \Rightarrow 3,25$$

$$i(t) \text{ for } t > 0$$

$$i(t) = 3,25 e^{-8t} + 0,75A$$

$$\tau = \frac{L}{R} = \frac{2}{16} = 0,125s$$

Sinusoidal Steady State:

$$\textcircled{1} \quad V_s(t) = 20 \sin(10t - \pi/3) \text{ V}$$

$$i_s(t) = 4 \cos(10t - \pi/4) \text{ A}$$

$$\omega = 10 \text{ rad/s}$$

$$Z_R = R = 10 \Omega$$

$$Z_L = j\omega L = j10 \Omega$$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j \cdot 10 \cdot 0.02} = -j5 \Omega$$

$$20 \sin(10t - \pi/3) = 20 \cos(10t - \pi/3 - \pi/2) = 20 \cos(10t - 5\pi/6)$$

$$V_s = 20 \angle -5\pi/6$$

$$i_s(t) = 4 \cos(10t - \pi/4) \quad I_s = 4 \angle -\pi/4$$

$$\frac{1}{Z_{LC}} = \frac{1}{Z_L} + \frac{1}{Z_C} \quad Z_{LC} = 10j \quad Z_{total} = Z_R + Z_{LC} = 10 + 10j$$

$$I_{total} = \frac{V_s}{Z_{total}} = \frac{20 \angle -5\pi/6}{10 + 10j} \quad |Z_{total}| = \sqrt{10^2 + 10^2} = 10\sqrt{2} \quad \theta_2 = \tan^{-1}(\frac{10}{10}) = \frac{\pi}{4}$$

$$Z_{total} = 10\sqrt{2} \angle \frac{\pi}{4} \quad I_{total} = \frac{20 \angle -5\pi/6}{10\sqrt{2} \angle \frac{\pi}{4}} \quad |I_{total}| = \sqrt{2} \angle I_{total} = \frac{13\pi}{12}$$

$$I_{total} = \sqrt{2} \angle -\frac{13\pi}{12} \text{ A} \quad \text{SO} \rightarrow i_o(t) = 2 \cos(10t - \frac{13\pi}{3}) \text{ A}$$

$$\textcircled{2} \quad V_1$$

$$\frac{V_1}{20} + \frac{V_1 - V_2}{40} + \frac{V_1 - V_2}{-j5} = j2$$

$$(0.15 + j0.2)V_1 - (0.1 + j0.2)V_2 = j2 \quad \dots (1)$$

$$V_2$$

$$\frac{V_2 - V_1}{10} + \frac{V_2 - V_1}{-j5} + \frac{V_2}{j10} = -(1 + j)$$

$$-(0.1 + j0.2)V_1 + (0.1 + j0.1)V_2 = -(1 + j) \quad \dots (2)$$

Matrix

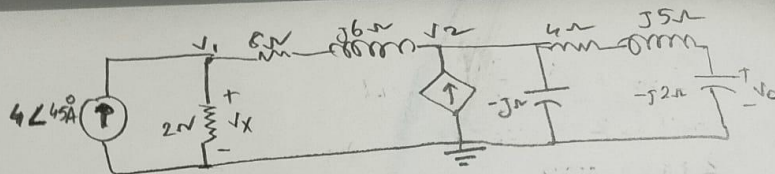
$$\begin{bmatrix} 0.15 + j0.2 & -(0.1 + j0.2) \\ -0.1 + j0.2 & 0.1 + j0.1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} j2 \\ -(1 + j) \end{bmatrix}$$

$$V = A^{-1} \cdot B$$

$$V_1 = -3.08 - j4.62$$

$$V_2 = -12.31 - j8.46$$

③



for node 1

$$4\angle 45^\circ = \frac{V_1}{2} + \frac{V_1 - V_2}{8 + j6}$$

$$200\angle 45^\circ = (29 - j3)V_1 - (4 - j3)V_2 \quad \text{--- (1)}$$

for node 2

$$2V_x + \frac{V_1 - V_2}{8 + j6} = \frac{V_2}{-j} + \frac{V_2}{j5 - j2 + 4}$$

$$V_1(104 - j3) = (12 + j41)V_2 \quad \text{--- (2)}$$

$$V_1 = \frac{12 + j41}{104 - j3} V_2 \quad \text{for } V_x = V_1$$

--- (2) with --- (1)

$$200\angle 45^\circ = (29 - j3) \frac{(12 + j41)}{104 - j3} V_2 - (4 - j3)V_2$$

$$200\angle 45^\circ = (14.21\angle 89.17^\circ)V_2$$

$$V_2 = \frac{200\angle 45^\circ}{14.21\angle 89.17^\circ}$$

$$V_0 = \frac{-6 - j8}{25} V_2$$

$$V_0 = \frac{10\angle 233.13^\circ}{25} \cdot \frac{200\angle 45^\circ}{14.21\angle 89.17^\circ}$$

$$V_0 = 5.63\angle 189^\circ \text{ V}$$

④

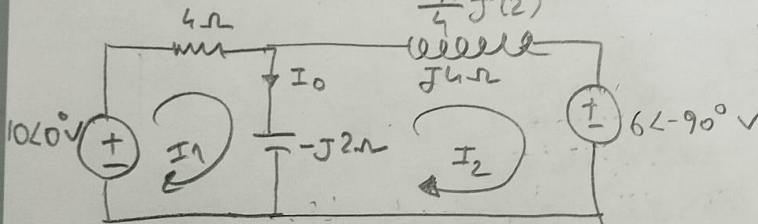
$$\omega = 2$$

$$10 \cos(2t) \Rightarrow 10 \angle 0^\circ$$

$$6 \sin(2t) \Rightarrow 6 \angle -90^\circ = -j6$$

$$2H \Rightarrow j\omega L = j4$$

$$0.25F \Rightarrow \frac{1}{j\omega C} = \frac{1}{\frac{1}{4}j(2)} = -j2$$



for 1

$$(4-j2)I_1 - 10 + j2I_2 = 0$$

$$5 = (2-j)I_1 + jI_2$$

--- (1)

for 2

$$j2I_1 + (j4-j2)I_2 + (-j6) = 0$$

$$I_1 + I_2 = 3$$

--- (2)

$$\begin{bmatrix} 2-j & j \\ 1 & 1 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$\Delta = 2(1-j) \quad \Delta_1 = 5-j3 \quad \Delta_2 = 1-j3$$

$$I_0 = I_1 - I_2 = \frac{\Delta_1 - \Delta_2}{\Delta} = \frac{4}{2(1-j)} = 1+j = 1.41 \angle 45^\circ$$

$$i_0(t) = 1.414 \cos(2t + 45^\circ) A$$

⑤ mesh 1

$$-40\angle 30^\circ + (j10 - j20)I_1 + j20I_2 = 0$$

$$4\angle 30^\circ = -jI_1 + j2I_2 \quad \text{--- (1)}$$

mesh 2

$$50\angle 0^\circ + (40 - j20)I_2 + j20I_1 = 0$$

$$5 = -j2I_1 - (4 - j2)I_2 \quad \text{--- (2)}$$

--- (1) with --- (2)

$$\begin{bmatrix} 4\angle 30^\circ \\ 5 \end{bmatrix} = \begin{bmatrix} -j & j2 \\ -j2 & -(4-j2) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

$$\Delta = -2 + 4j = 4.472 \angle 116.56^\circ$$

$$\Delta_1 = -(4\angle 30^\circ)(4-j2) - j10 = 21.01 \angle 211.8^\circ$$

$$\Delta_2 = -j5 + 8\angle 120^\circ = 4.44 \angle 154.27^\circ$$

$$I_1 = \frac{\Delta_1}{\Delta} = 4.698 \angle 95.24^\circ \text{ A}$$

$$I_2 = \frac{\Delta_2}{\Delta} = 0.9928 \angle 37.7^\circ \text{ A}$$

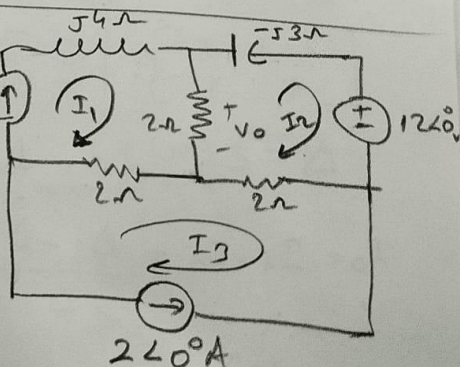
⑥ $I_1 = 4\angle 90^\circ = j4$ and $I_3 = -2\text{ A}$

mesh 2

$$(4-j3)I_2 - 2I_1 - 2I_3 + 12 = 0$$

$$(4-j3)I_2 - j8 + 4 + 12 = 0$$

$$I_2 = \frac{j8-16}{4-j3} = -3.52 - j0.64$$



$$V_0 = 2(I_1 - I_2) = 2(3.52 + j4.64) = 7.04 + j9.28$$

$$V_0 = 11.648 \angle 57.82^\circ \text{ V}$$

⑦ $V_0 = V_1 + V_2 + V_3$

will be $V_1 = 10V$

for V_2 ;

$\omega = 2$

$2H \rightarrow j\omega L = j4$

$\frac{1}{12}F \rightarrow \frac{1}{j\omega C} = \frac{1}{j(2) \cdot (1/12)} = -j6$

nodal analysis

$4 = \frac{V_2}{6} + \frac{V_2}{-j6} + \frac{V_2}{j4} = \left(\frac{1}{6} + \frac{j}{6} - \frac{j}{4}\right)V_2$

$V_2 = \frac{24}{1 - j0.5} = 21.45 \angle 26.56^\circ$

$V_2 = 21.45 \sin(2t + 26.56^\circ) V$

for V_3

$\omega = 3$

$2H \rightarrow j\omega L = j6$

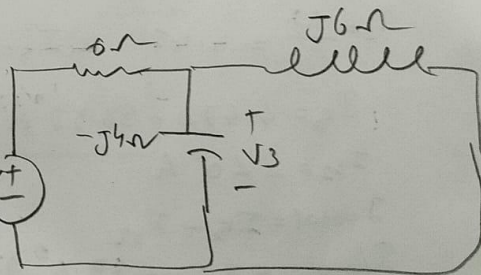
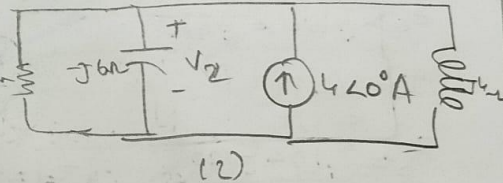
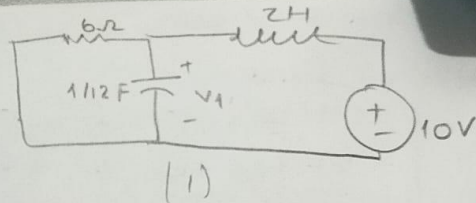
$\frac{1}{12}F \rightarrow \frac{1}{j\omega C} = \frac{1}{j(3) \cdot \frac{1}{12}} = -j4$

non-reference node

$\frac{12 - V_3}{6} = \frac{V_3}{j6} + \frac{V_3}{-j4}$

$V_3 = \frac{12}{1 + j0.5} = 10.73 \angle -26.56^\circ$

$V_3 = 10.73 \cos(3t - 26.56^\circ) V$



$V_0 = 10 + 21.45 \sin(2t + 26.56^\circ) + 10.73 \cos(3t - 26.56^\circ) V$

⑧ sinusoidal V source $= 10 \sin(3t - 30^\circ) \text{ V}$

(a) Voltage Source

$$10 \sin(3t - 30^\circ) = 10 \cos(3t - 30^\circ - 90^\circ) = 10 \cos(3t - 120^\circ)$$

$$V_s = 10 \angle -120^\circ$$

(b)

$$Z_c = \frac{1}{j\omega C} = \frac{1}{j \cdot 3 \cdot (1/6)} = -j2 \Omega$$

$$Z_{1c} = 1 - j2 \Omega$$

(c)

$$Z_L = j\omega L = j6 \Omega$$

$$I = \frac{V}{Z}$$

$$I_s = \frac{10 \angle -120^\circ}{1 - j2}$$

$$|Z| = \sqrt{1^2 + (-2)^2} = \sqrt{5}$$

phase of Z

$$\theta_Z = \tan^{-1}\left(\frac{-2}{1}\right) = -63.43^\circ$$

$$Z = \sqrt{5} \angle -63.43^\circ$$

$$I_s = \frac{10}{\sqrt{5}} = 4.47 \quad \angle I_s = -120^\circ + 63.43^\circ = -56.57^\circ$$

$$I_s = 4.47 \angle -56.57^\circ \text{ A}$$

$$I_{s2} = 2 \angle 0^\circ \text{ A}$$

$$I_{\text{total}} = I_s + I_{s2}$$

$$I_{\text{total}} = 5.8 \angle -39.6^\circ \text{ A}$$

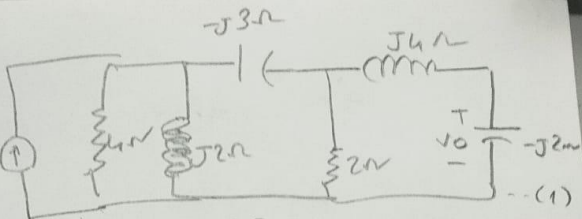
$$i_o(t) = 5.8 \cos(3t - 39.6^\circ) \text{ A}$$



$$Z_s = 4 \parallel j2 = \frac{j8}{4+j2} = 0.8 + j1.6$$

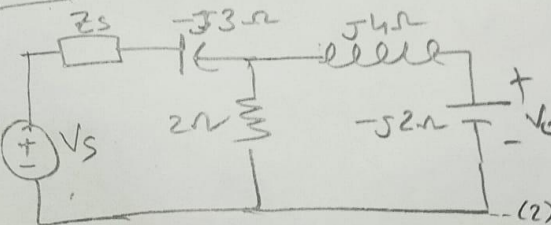
$$V_s = (5 \angle 0^\circ) Z_s = 5(0.8 + j1.6) = 4 + j8$$

$$V_s = 4 + j8$$



$$Z_x = Z_s - j3 = 0.8 - j1.4$$

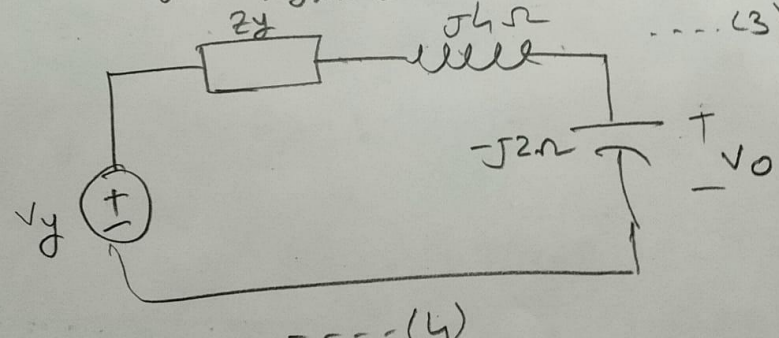
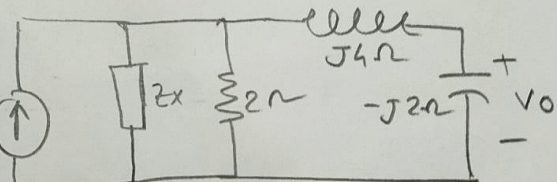
$$I_x = \frac{V_s}{Z_s} = \frac{4 + j8}{0.8 - j1.4} = -3.077 + j4.6154$$



--- (1) with --- (2)

$$Z_y = 2 \parallel Z_x = \frac{1.6 - j2.8}{2.8 - j1.4} = 0.8571 - j0.5714$$

$$V_y = I_x Z_y = (-3.0769 + j4.6154) \times (0.8571 - j0.5714) = j5.7143$$



$$V_o = \frac{-j2}{Z_y + j4 - j2} V_y = \frac{-j2}{0.8571 - j0.5714 + j4 - j2} (j5.7143)$$

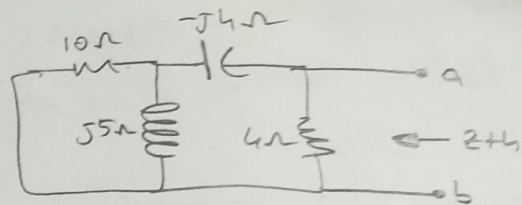
$$V_o = (3.529 - j5.883) V$$

(10) z_{th}

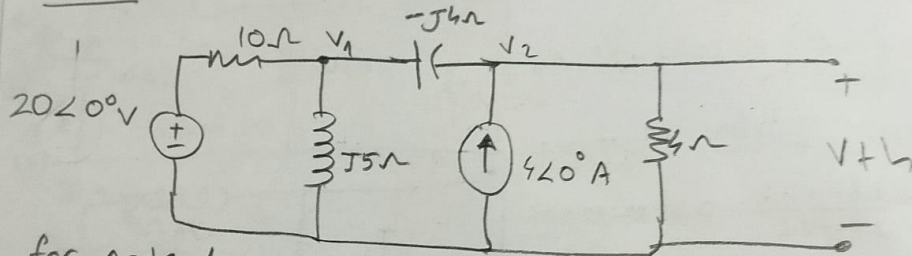
$$z_{th} = 4 \parallel (-j4 + 10 \parallel j5)$$

$$z_{th} = 4 \parallel (-j4 + 2 + j4)$$

$$z_{th} = 4 \parallel 2 = 1.333 \Omega$$



V_{th}



for node 1

$$\frac{20 - V_1}{10} = \frac{V_1}{j5} + \frac{V_1 - V_2}{-j4} \quad (1 + j0.5)V_1 - j2.5V_2 = 20$$

for node 2

$$4 + \frac{V_1 - V_2}{-j4} = \frac{V_2}{4} \quad V_1 = (1 - j)V_2 + j16$$

---(2) with ---(1)

$$28 - j16 = (1.5 - j3)V_2$$

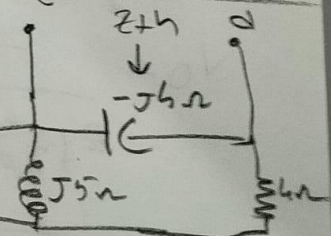
$$V_2 = \frac{28 - j16}{1.5 - j3} = 8 + j5.333$$

$$V_{th} = V_2 = 9.615 \angle 33.69^\circ V$$

z_{th}

$$z_{th} = -j4 \parallel (4 + 10 \parallel j5) = -j4 \parallel (4 + \frac{j10}{2+j})$$

$$z_{th} = -j4 \parallel (6 + j4) = \frac{-j4}{6} (6 + j4) = 2.667 - j4 \Omega$$



$$V_2 = 8 + j5.333 = (8/3) (3 + j2)$$

$$V_1 = (1 - j)V_2 + j16 = j16 + (8/3)(5 - j)$$

$$V_{th} = V_1 - V_2 = 16/3 + j8 = 9.614 \angle 56.31^\circ V$$